Experiment No. 2

Experiment Name

Exploratory Data Analysis (EDA) on a Real-World Business Dataset using Python

Aim

To perform Exploratory Data Analysis (EDA) on a real-world dataset using Python in order to understand the dataset's structure, identify trends, detect anomalies, analyze feature relationships, and generate meaningful business insights through descriptive statistics and visualizations.

Purpose

Exploratory Data Analysis (EDA) is one of the most critical stages in the data analytics lifecycle. Before building predictive models or making business decisions, analysts must thoroughly understand the data. EDA helps identify data distributions, relationships between variables, hidden patterns, anomalies, and potential issues such as skewness or class imbalance.

In this experiment, students will analyze a real-world dataset such as the Netflix Movies and TV Shows Dataset, Superstore Sales Dataset, or IBM HR Employee Attrition Dataset. Using statistical summaries and visualization techniques, students will derive actionable insights that can support data-driven decision-making.

Prerequisites

- Basic knowledge of Python programming
- Familiarity with Pandas, NumPy, Matplotlib, and Seaborn
- Understanding of descriptive statistics (mean, median, mode, variance, standard deviation)
- Completion of data preprocessing (Experiment 1)
- Jupyter Notebook or Google Colab

Steps

1. Select a real-world dataset such as the Netflix Titles Dataset, Superstore Sales Dataset, or IBM HR Employee Attrition Dataset.
2. Import the required Python libraries (Pandas, NumPy, Matplotlib, and Seaborn) and load the cleaned dataset.
3. Explore the dataset by examining its dimensions, data types, column names, and summary statistics using functions such as info(), describe(), and value_counts().

4. Analyze the distribution of numerical features using histograms and density plots to understand central tendency and data spread.
5. Visualize categorical variables using bar charts and count plots to identify the most frequent categories.
6. Generate a correlation matrix and heatmap to identify relationships between numerical variables and detect highly correlated features.
7. Create box plots and scatter plots to identify outliers and study relationships between different variables.
8. Perform univariate, bivariate, and multivariate analysis to uncover trends and business patterns within the dataset.
9. Summarize key observations and identify factors that significantly influence the target variable or business objective.
10. Prepare a concise report highlighting important findings and recommendations based on the exploratory analysis.

Questions

1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?
2. Differentiate between univariate, bivariate, and multivariate analysis with suitable examples.
3. What insights can be obtained from a correlation heatmap?
4. Explain the purpose of histograms, box plots, and scatter plots in EDA.
5. How can EDA help identify data quality issues before analysis?
6. Why is correlation important in predictive analytics? Can correlation imply causation?
7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.
8. What business insights can be derived from the Netflix (or Superstore/HR Analytics) dataset through EDA?
9. How does EDA contribute to feature selection and model building?
10. What challenges might arise while performing EDA on large-scale real-world datasets?